Tim Richardson

Senior Software Engineer • Engineering Manager • Genomics and AI

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Summary

Senior software engineer and engineering manager with 13+ years building scalable data platforms, specialising in genomics and healthcare. Currently leading AI transformation and managing an engineering team at Genomics England. Track record delivering full-stack systems in Python, Go and TypeScript on AWS and GCP, with deep experience of bioinformatics pipelines and ML integration. Builds and publishes open-source tooling for AI coding agents, tested with pre-registered experiments.

Technical Skills

Languages	Python, Go, TypeScript, JavaScript, Rust, SQL
AI and ML	LLMs, RAG, LangChain, OpenAI and Anthropic APIs, Google ADK, NumPy, SciPy, Pandas
AI agents	Claude Code plugins and skills, agent memory, agent evaluation and benchmarking
Testing	pytest, mutation testing, coverage analysis, TDD, CI quality gates
Backend	FastAPI, Django, Node.js, GraphQL, REST, microservices
Frontend	React, Next.js, Redux, Svelte, Tauri, D3.js
Data	PostgreSQL, Elasticsearch, Redis, vector and graph databases, FHIR, ontologies
Cloud	AWS, GCP, Docker, Kubernetes, Terraform, CI/CD
Bioinformatics	Nextflow, nf-core, VEP, Samtools, BWA, Singularity, LSF

Experience

Genomics England — Dec 2021 – Present
Senior Software Engineer and Performance Development Lead — London, UK

- **Engineering leadership:** line manager for 6 direct reports, responsible for mentoring, performance management and delivery.
- **AI transformation:** leading the internal programme that introduces AI coding assistants across engineering.
- **Workflow modernisation:** key role in migrating complex workflow orchestration to Nextflow, improving scalability and reproducibility.
- **Backend development:** designing and maintaining Python services on AWS for high-throughput genomic data processing.

IQVIA — Mar 2019 – Dec 2021
Senior Software Engineer, AI/ML Team — London, UK
Go, FastAPI, TypeScript, React, Redux, Elasticsearch, PostgreSQL, D3.js

- Built a data analytics product for real-world healthcare data: Go backend services and a React frontend.
- Deployed machine learning models with FastAPI for scalable inference in healthcare analytics.
- Designed and implemented the product's statistical features.
- Coordinated releases, product development, process improvement and code review.

Project Shivom — Oct 2018 – Mar 2019
Lead Developer — London, UK
Blockchain, smart contracts, Google Genomics, Cloud Build, PostgreSQL, GraphQL, React Apollo

- Designed, built and deployed a genomic data sharing platform with blockchain components.
- Managed the development team; built the REST backend and React frontend; deployed with Docker and App Engine.

Congenica — Mar 2016 – Jul 2018
Senior Frontend Developer — Hinxton, UK
Perl, React, Redux, reactive programming, BioPerl

- Developed and optimised features of a clinical genomics tool with Perl APIs and a React frontend, focusing on data flow in a data-intensive UI.
- Led development of the API integrating Congenica's Sapientia platform with Genomics England's 100,000 Genomes Project.

Repositive — Nov 2014 – Mar 2016

Lead Developer London, UK

Docker, Node.js, Ansible, Ember.js, PostgreSQL, Elasticsearch

- Led development of a genomic data discovery platform, including its infrastructure and development process.
- Designed the Node.js and Express API on Redis, PostgreSQL and Elasticsearch, with an Ember.js single-page frontend.

Earlier roles

Polecat, Python Contractor	Python reporting service generating PowerPoint reports (Django, GraphQL, React, Elasticsearch)	2018
Wolfson Institute, Freelance	ChIP-seq analysis: QC, peak calling, motif discovery, GO annotation	2014
Fire Tech Camp, Instructor	Taught Python to children aged 11 to 15	2014
Ixxy, Django Developer	Sites on a Django CMS	2013 – 2014
Ecomarket.com, Django Developer	Backend for an online marketplace (Django, PostgreSQL)	2013
Swerve Concepts, Junior Developer	Full-stack Python and PHP; built Swerve Media Hub and SignUpAnywhere	2011 – 2013

Selected Open Source

Agentic Verification github.com/timini/agentic-verification

Claude Code plugin and GitHub Action that checks whether tests catch broken code by running every mutant of the lines a change touches. Backed by a pre-registered study and agent benchmarks on 8 open-source Python libraries, published as a paper. On 14 real commits, the agent's tests caught 99% of mutants with the plugin against 92% without.

drill-me github.com/timini/drill-me

Claude Code plugin that turns Claude into an adaptive tutor: it quizzes before it explains, schedules reviews with spaced repetition and remembers what you got wrong across sessions.

neuromem github.com/timini/neuromem

Long-term memory for LLM agents modelled on hippocampal consolidation, with providers for Gemini, OpenAI, Claude and local models. Scores 0.870 on the LongMemEval-s benchmark (n=100).

Education

Birkbeck, University of London 2012 – 2014

MSc Bioinformatics and Systems Biology (Merit) London, UK

Statistics, protein structure, sequence alignment, systems biology and software development. Thesis: tracing human migration with MCMC population clustering of dense genotype data.

University of Leeds 2005 – 2008

BSc Cognitive Science (Hons) Leeds, UK

Computing, psychology and philosophy. Dissertation: data mining techniques for dynamic data sets.